

Anatomy of the Electronic Prescription and Machine-Learning Phenotyping of a Community Telemedicine Cohort: Evidence from 46,000 Portable Health Clinic Encounters in Rural Bangladesh

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Abstract—Community telemedicine programs generate large, real-world screening datasets whose prescription and phenotype structure remains under-analyzed. We conduct a computational study of 46,056 Portable Health Clinic (PHC) encounters (36,530 patients) collected by community health workers across seven divisions of Bangladesh. We (i) decompose 30,016 electronic prescription drug lines into the four classical prescription components (superscription, inscription, subscription, signatura); (ii) quantify concordance between prescribing and measured vitals; (iii) mine drug co-prescription association rules; (iv) derive unsupervised metabolic phenotypes; and (v) benchmark regularized logistic regression, gradient-boosted trees, and a PyTorch multilayer perceptron for predicting hypertension and dysglycemia from routine vitals. Prescriptions average 2.74 drugs (11.7% polypharmacy ≥ 5), are 86.5% tablets, and 64.9% carry meal-timing instructions. Prescribing tracks measurement tightly (antihypertensive $\rightarrow$ mean systolic 143.7 vs. 120.9 mmHg), yet 49.2% of measured-hypertensive prescribed encounters received no antihypertensive—a care gap that narrows with severity. Four reproducible phenotypes emerge, and gradient boosting attains ROC-AUC 0.78–0.81, statistically matched by the deep model. All findings are hypothesis-generating and framed against program-population selection bias.

Index Terms—telemedicine, electronic prescription, machine learning, hypertension, association rule mining, global health, Bangladesh

I. Introduction

Non-communicable diseases (NCDs) such as hypertension and diabetes are rising rapidly in South Asia, while access to physicians in rural areas remains limited. The Portable Health Clinic (PHC), a telemedicine system developed for unreached communities in Bangladesh, addresses this gap: trained community health workers visit patients with a portable diagnostic kit, record vitals and symptoms on-site, and transmit them to a remote physician who returns a diagnosis and an electronic prescription [1]–[3]. Over years of operation the program has accumulated a large corpus of structured screening encounters that is well suited to computational analysis but is rarely mined beyond operational reporting.

This paper treats the PHC record as a data-science object. We ask three questions. First, what is the structure of the electronic prescription—its size, dosage forms, dosing frequencies, and instruction completeness—when decomposed into the four pharmacologically classical components? Second, does prescribing behavior concord with the measured vitals captured at the same encounter, and where are the care gaps? Third, can unsupervised and supervised machine learning extract clinically coherent patient phenotypes and predict NCD status from routine vitals, and do modern deep tabular models outperform classical baselines?

Our contributions are: (1) a reproducible decomposition of 30,016 prescription drug lines into superscription/inscription/subscription/signatura with completeness auditing; (2) a prescribing-versus-measurement concordance analysis that surfaces a quantified, severity-dependent hypertension treatment gap; (3) co-prescription association-rule mining and a therapeutic-class network; (4) unsupervised metabolic phenotyping; and (5) a leakage-controlled benchmark of logistic regression, gradient boosting, and a deep MLP, with SHAP-based interpretation. Throughout, we frame results as hypotheses subject to the selection biases inherent to a screening program.

II. Related Work

Portable Health Clinic. The PHC system and its “GramHealth” data backbone were introduced to deliver preventive care to unreached communities using appropriate, low-cost sensing [1], [3]. A verification study of the health-checkup and telemedical-intervention program demonstrated measurable risk-stratification and follow-up value in developing-country settings [4], and later work advanced the platform toward broader tele-healthcare deployment [2]. These studies establish the system and its clinical rationale; our work complements them by mining

the accumulated prescription and vitals records at scale rather than evaluating the intervention itself.

Hypertension epidemiology in Bangladesh. Nationally representative analyses of the Bangladesh Demographic and Health Survey report adult hypertension prevalence near 27% with low awareness, treatment, and control rates [5], [6], and systematic reviews document a rising secular trend [7], [8]. Our cohort is a program population rather than a general-population sample, so we use these works as external reference points, not as directly comparable denominators.

Machine learning on tabular clinical data. Gradient-boosted decision trees remain a dominant approach for heterogeneous tabular data [9], and recent controlled benchmarks show tree ensembles frequently match or exceed deep networks on typical tabular problems [10]. We adopt SHAP for model-agnostic interpretation [11], Apriori for association-rule mining [12], and standard scikit-learn and PyTorch implementations [13], [14]. To our knowledge this is the first study to apply this combined prescription-anatomy and ML pipeline to a PHC-scale telemedicine cohort.

III. Dataset and Methods

Data source. We analyzed a de-identified operational database export (135 tables) from the PHC program comprising 46,056 checkup encounters for 36,530 patients across seven administrative divisions. Each encounter records patient demographics, on-site vitals (blood pressure, pulse, SpO₂, random/post-breakfast glucose, height, weight/BMI, and sparsely hemoglobin), and, where issued, a structured electronic prescription. Prescription drug lines (30,016) link to encounters through the prescription-checkup-patient key chain.

Units and reference standards. Glucose was 98.4% mg/dL with a 1.3% mmol/L cluster harmonized by $\times 18$. Blood pressure is in mmHg. Hypertension was defined as systolic ≥ 140 or diastolic ≥ 90 mmHg (WHO), and dysglycemia by an elevated glucose threshold consistent with post-breakfast testing, the dominant assay. Physiologically implausible extremes were removed with fixed plausibility ranges before analysis; missingness was reported rather than silently dropped.

Prescription anatomy. Each drug line was mapped to the four classical components: superscription (prescription-level drug count), inscription (drug name and therapeutic class; 206 high-frequency brands mapped to generics/classes covering 79.8% of volume, remainder flagged unknown), subscription (dosage form and duration), and signatura (dose frequency from day-part codes and meal-timing).

Concordance. For prescribed encounters we compared mean measured vitals by whether the matching therapeutic class was given, and computed the fraction of measured-hypertensive encounters receiving no antihypertensive, stratified by severity.

TABLE I: Prescription-component summary and field completeness

Component / field	Value
Drug lines / prescriptions	30,016 / 10,952
Mean drugs per prescription	2.74 (median 2)
Polypharmacy (≥ 5 drugs)	11.7%
Tablet dosage form	86.5%
Once-daily dosing	47.2%
After-meal instruction	64.9%
Completeness: drug name	99.6%
Completeness: dosage form	98.7%
Completeness: dose	93.7%
Completeness: meal-timing	74.6%

Association rules. Apriori [12] was applied to 10,299 prescription baskets at both therapeutic-class and brand granularity (minimum support 0.01), ranked by lift.

Phenotyping. k -means ($k=4$) was fit on seven standardized first-visit vitals for the 24,381 complete-core patients (hemoglobin dropped, 79.4% missing); PCA and UMAP provided projections. $k=4$ was chosen for clinical interpretability although silhouette favored $k=2$.

Predictive modeling. For hypertension and dysglycemia we trained regularized logistic regression and histogram gradient boosting, excluding the defining vital of each outcome to prevent label leakage. Data were split 75/25 stratified (seed 42); we report ROC-AUC, PR-AUC, and cross-validated AUC, with SHAP [11] for interpretation. A PyTorch MLP (64 $\rightarrow$ 32, batch-norm, dropout 0.3, positive-weighted BCE, early stopping on validation AUC) was trained on CPU as the deep-learning benchmark [14].

IV. Results

A. Prescription anatomy

The 30,016 drug lines span 10,952 prescriptions (99.7% linked to an encounter) and 1,367 distinct brands. Superscription: a mean of 2.74 drugs per prescription (median 2), with 11.7% polypharmacy (≥ 5 drugs). Inscription: proton-pump inhibitors (PPIs), vitamins/minerals, and antibiotics dominate (Fig. 1c). Subscription: 86.5% of lines are tablets and 48.2% carry a month-or-continuous duration. Signatura: 47.2% once-daily, 10.7% three-times-daily, and 64.9% carry an after-meal instruction. Field completeness was high (drug name 99.6%, form 98.7%, dose 93.7%) but meal-timing was present in only 74.6% of lines (Table I), the principal documentation gap.

B. Prescribing-measurement concordance and care gap

Prescribing tracked measured physiology tightly. Encounters receiving an antihypertensive had a mean systolic BP of 143.7 mmHg versus 120.9 mmHg for those that did not; antidiabetic prescriptions corresponded to mean glucose 272.1 versus 116.0 mg/dL (Fig. 2b). This internal consistency supports data integrity. Nonetheless, among measured-hypertensive prescribed encounters, 49.2% received no antihypertensive; the gap narrowed with severity, from 49.9% at Stage-1 (140–159 mmHg) to 28.6% at

Anatomy of the PHC electronic prescription (30,016 drug lines, 10,952 prescriptions)

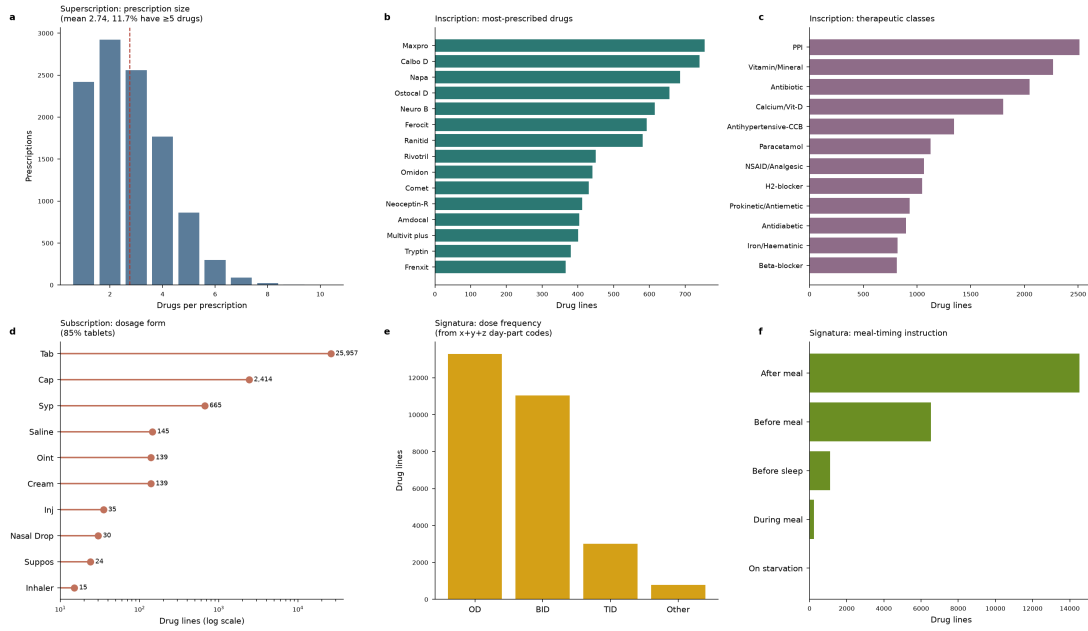


Fig. 1: Anatomy of the PHC electronic prescription (30,016 drug lines, 10,952 prescriptions). (a) Superscription: drugs per prescription. (b-c) Inscription: most-prescribed drugs and therapeutic classes. (d) Subscription: dosage form (log scale). (e-f) Signatura: dose frequency and meal-timing instructions.

Therapeutic-class prescribing and concordance with measured vitals — PHC

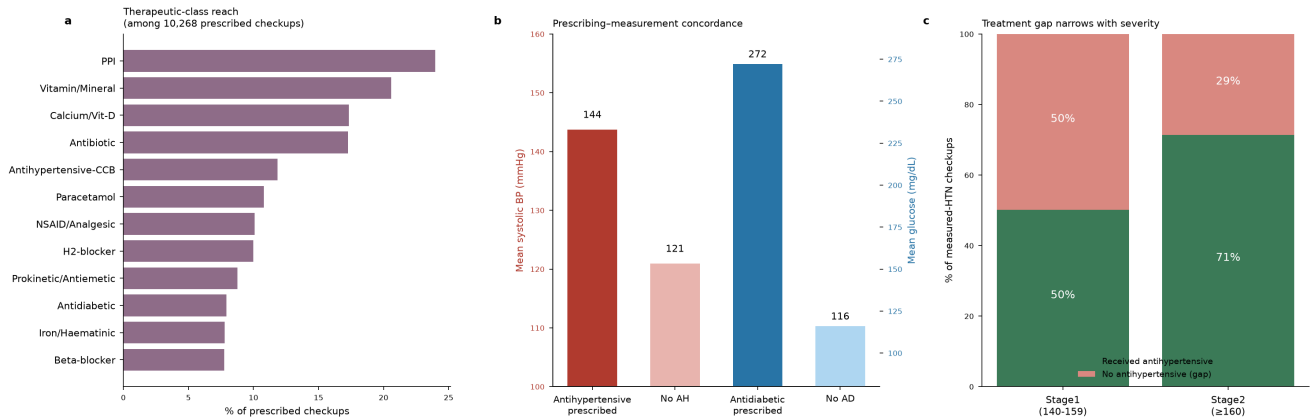


Fig. 2: Therapeutic-class prescribing and concordance with measured vitals. (a) Class reach among 10,268 prescribed encounters. (b) Prescribing-measurement concordance: mean systolic BP and glucose by whether the matching class was prescribed. (c) Antihypertensive treatment gap narrows from Stage-1 to Stage-2 hypertension.

Stage-2 (≥ 160 mmHg) (Fig. 2c). A comparable 46.9% gap was observed for dysglycemia. The denominator is restricted to encounters with a prescription record, so the absolute gap should be read as within-prescribed-encounters rather than population-level under-treatment.

C. Drug co-prescription association rules

Apriori mining of 10,299 baskets yielded 17 therapeutic-class and 18 brand-level rules. The strongest class associations centered on gastro-protective and musculoskeletal co-prescribing: Calcium/Vit-D + PPI $\rightarrow$ NSAID (confidence 0.25, lift 2.43 $\times$); NSAID $\leftrightarrow$ H2-blocker (0.22, 2.19 $\times$); NSAID $\leftrightarrow$ Calcium/Vit-D (0.36, 2.08 $\times$); and

Antacid $\rightarrow$ PPI (0.37, 1.56 $\times$). In the therapeutic-class network (Fig. 3), the PPI node is the central hub, consistent with routine gastro-protective co-prescription alongside analgesics and supplements.

D. Unsupervised metabolic phenotyping

Four reproducible phenotypes emerged (Fig. 4): Young-lean (42.9%; age 32, BMI 20, BP 108/70), Younger-high-BMI (35.0%; age 32, BMI 25, BP 122/82), Older-hypertensive (18.2%; age 55, BP 144/88), and Dysglycemic (4.0%; glucose 307, age 49). PCA's first component (31.6% variance) is a blood-pressure/age axis; the dysglycemic group separates along the glucose direction. Although

Co-prescription network of therapeutic classes
(edge width $\propto$ lift; node size $\propto$ prescribing frequency; labels = lift for top pairs)



Fig. 3: Therapeutic-class co-prescription network (edge width $\propto$ lift, node size $\propto$ frequency). PPI is the central hub.

TABLE II: Predictive performance (test $n \approx 8,100$; ROC-AUC)

Outcome	Logistic	Grad. Boost	Deep MLP
Hypertension	0.769	0.780	0.778
Dysglycemia	0.746	0.805	0.799

silhouette favored $k=2$, the four-cluster solution is clinically more actionable, isolating the older-hypertensive and dysglycemic risk groups that dominate the prescribing signal.

E. Predictive modeling and deep learning

With the defining vital excluded to prevent leakage, models predicted NCD status from routine vitals at clinically useful discrimination (Table II; Fig. 5). For hypertension, gradient boosting reached ROC-AUC 0.780 (PR-AUC 0.501) versus logistic 0.769; for dysglycemia, 0.805 versus 0.746. SHAP attributed dominant importance to age, followed by BMI, pulse, and SpO₂. Calibration curves lay below the diagonal, an expected consequence of class-weighting that inflates predicted probabilities; probabilities should be recalibrated before any operational use.

The PyTorch MLP matched the tree ensemble rather than beating it (hypertension AUC 0.778, dysglycemia 0.799; Fig. 6), consistent with reports that tree models remain strong on tabular data [10]. A regression head predicting polypharmacy count from vitals was essentially uninformative ($R^2=0.015$, MAE 1.15 drugs)—an honest negative indicating that prescription burden is not recoverable from vitals alone.

V. Discussion

Three findings stand out. First, the electronic prescription is highly structured and largely complete, making it a viable substrate for automated pharmacovigilance and care-quality auditing; meal-timing documentation (74.6%) is the clearest target for form-level improvement. Second, prescribing concordance with measured vitals is strong, yet

a substantial within-encounter treatment gap persists for hypertension and dysglycemia and narrows with severity—a pattern consistent with clinicians prioritizing the most abnormal cases, and a candidate target for decision support. Third, routine vitals carry real predictive signal for NCD status (AUC 0.78–0.81), but deep learning offers no advantage over gradient boosting on this tabular problem, echoing the broader literature [10].

These results align directionally with national evidence of high, under-treated hypertension in Bangladesh [5], [7], suggesting operational telemedicine data can serve as a continuously updating epidemiological sensor.

VI. Limitations

This is a program (worksite/camp) population, not a general-population sample, so prevalences are not directly comparable to survey denominators and are subject to self-selection into screening. The design is cross-sectional and single-visit, precluding causal inference. Age is year-precision; hemoglobin and cholesterol are sparse; brand-to-generic mapping covers 79.8% of volume with an unknown tail; and the care-gap denominator is restricted to prescribed encounters. Models discriminate well but are uncalibrated under class-weighting, and the GPU was unavailable (CPU-only training). All findings are hypothesis-generating and require prospective, adjusted confirmation.

VII. Conclusion

Treating a community telemedicine archive as a data-science object, we decomposed 30,016 electronic prescriptions, quantified a severity-dependent hypertension treatment gap, mined co-prescription structure, derived four metabolic phenotypes, and benchmarked classical and deep predictors of NCD status. The pipeline turns routinely captured PHC records into actionable care-quality and epidemiological hypotheses, and is directly reusable as new encounters accrue. Future work should add longitudinal follow-up, confounder-adjusted association testing, probability recalibration, and prospective evaluation of a treatment-gap decision-support prompt.

Reproducibility and Ethics

Analyses used de-identified operational data. Figures and tables were produced by a scripted, seed-fixed pipeline; prescription mapping, model configurations, and plausibility filters are archived as artifacts to support replication.

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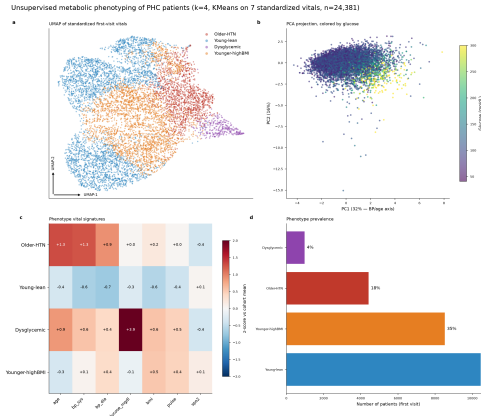


Fig. 4: Unsupervised metabolic phenotyping (k -means, $k=4$, on 7 standardized first-visit vitals, $n=24,381$). (a) UMAP; (b) PCA colored by glucose; (c) phenotype vital signatures (z -scores); (d) phenotype prevalence.

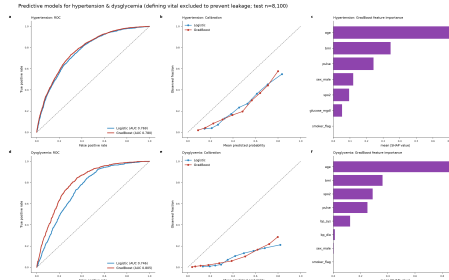


Fig. 5: Predictive models for hypertension and dysglycemia (defining vital excluded; test $n \approx 8,100$). ROC (a,d), calibration (b,e), and gradient-boosting SHAP importance (c,f).

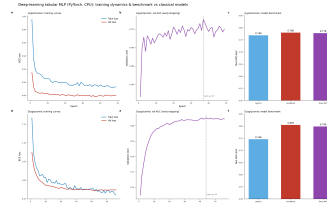


Fig. 6: Deep tabular MLP (PyTorch, CPU): training/validation curves (a,b,d,e) and benchmark vs classical models (c,f).

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